WORKED SOLUTION

STEP III 1993, Question 11

Mechanics • Moment of inertia

STEP III 1993, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q11

The key idea

Put coordinates on the cube and write every force as a vector. The six forces add up to zero, so all they do is apply a *couple*, and a couple has the same moment about every point. Part (i) is then about the direction of that moment. Part (ii) uses only its component along OO', together with a moment of inertia that we build up rod by rod using the distance from a point to a line.

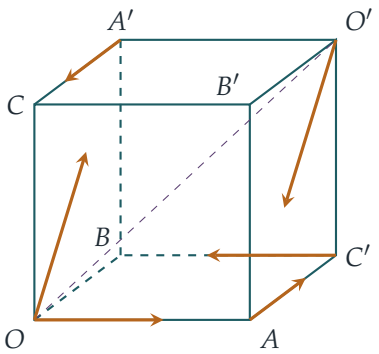
Solution

We model the frame as rigid, made of thin uniform rods, with no gravity. Take O as origin and unit vectors along OA, OB, OC, so that

$$A = (a, 0, 0), \quad B = (0, a, 0), \quad C = (0, 0, a),$$

and the opposite vertices are

$$O' = (a, a, a), \quad A' = (0, a, a), \quad B' = (a, 0, a), \quad C' = (a, a, 0).$$



The amber arrows show the six lines of action, and the diagonal OO' is dashed in plum.

Part (i)

Each force is its magnitude times the unit vector along its line. For example, $O'A$ is the vector $(0, -a, -a)$, of length $\sqrt{2}\,a$, so the force $\sqrt{2}\,mg$ along it is $mg(0, -1, -1)$. Doing the same for all six:

line	point on line	force
OA	O	$mg(2, 0, 0)$
AC'	A	$mg(0, 1, 0)$
$O'A$	A	$mg(0, -1, -1)$
OA'	O	$mg(0, 1, 1)$
$C'B$	B	$mg(-2, 0, 0)$
$A'C$	C	$mg(0, -1, 0)$

The forces sum to $\mathbf{0}$. A system with zero resultant is equivalent to a couple, whose moment is the same about every point. Taking moments about O (the forces along OA and OA' pass through O and contribute nothing):

$$\begin{aligned}\mathbf{G} &= A \times mg(0,1,0) + A \times mg(0,-1,-1) + B \times mg(-2,0,0) + C \times mg(0,-1,0)\\ &= amg[(0,0,1) + (0,1,-1) + (0,0,2) + (1,0,0)]\\ &= amg\,(1,1,2).\end{aligned}$$

The pivot at O can push on the frame, but its force acts through O and so has no moment about O. Starting from rest, the frame therefore begins to turn about the line through O along the axis of the couple, which has direction

$$\begin{pmatrix}1\\1\\2\end{pmatrix},$$

as required.

A note for the curious. This is the answer the question intends, and it's what you should write. Strictly, though, for a body free to turn in three dimensions the angular acceleration is $\mathbf{I}^{-1}\mathbf{G}$, where $\mathbf{I}$ is the inertia *tensor* about O. This points along $\mathbf{G}$ only when $\mathbf{G}$ lies along a principal axis. For this frame, about O, the moment of inertia is $14ma^2/3$ about OO' (from part (ii)) and $41ma^2/3$ about every axis through O perpendicular to OO'. This gives an initial axis in direction $(25, 25, 32)$, about 12.6° away from $(1, 1, 2)$.

Part (ii)

Let $\mathbf{u} = \frac{1}{\sqrt{3}}(1,1,1)$ be the unit vector along OO'. A point $\mathbf{p}$ is at squared distance

$$d^2 = |\mathbf{p}|^2 - (\mathbf{p}\cdot\mathbf{u})^2$$

from the line OO', by Pythagoras, since $\mathbf{p}\cdot\mathbf{u}$ is the component of $\mathbf{p}$ along the line.

Rod OA**.** The point at distance x from O is $(x, 0, 0)$, so $d^2 = x^2 - \frac{x^2}{3} = \frac{2}{3}x^2$. An element of length $\mathrm{d}x$ has mass $\frac{m}{a}\,\mathrm{d}x$, so

$$I_{OA} = \int_0^a \frac{m}{a}\cdot\frac{2x^2}{3}\,\mathrm{d}x = \frac{2ma^2}{9}.$$

The midpoint $(\frac{a}{2}, 0, 0)$ is at squared distance $\frac{2}{3} \cdot \frac{a^2}{4} = \frac{a^2}{6}$ from OO'. By the parallel axis theorem, about the parallel axis through the midpoint,

$$I_{\text{mid}} = \frac{2ma^2}{9} - m \cdot \frac{a^2}{6} = \frac{ma^2}{18}.$$

Rod $B'C$. This rod runs from $(a, 0, a)$ to $(0, 0, a)$, so it is parallel to OA. It is also a rod of the same length, placed at the same angle to the axis. So about the parallel axis through its own midpoint its moment of inertia is again $ma^2/18$. Its midpoint is $\mathbf{p} = (\frac{a}{2}, 0, a)$, with

$$d^2 = \frac{a^2}{4} + a^2 - \frac{1}{3}\left(\frac{3a}{2}\right)^2 = \frac{5a^2}{4} - \frac{3a^2}{4} = \frac{a^2}{2}.$$

By the parallel axis theorem,

$$I_{B'C} = \frac{ma^2}{18} + \frac{ma^2}{2} = \frac{5ma^2}{9}.$$

Ask yourself Do I really have to do this for all twelve rods?

No: symmetry does it for us. A rotation of 120° about OO' (cycling $x \to y \to z \to x$) and the point reflection in the centre of the cube, $\mathbf{p} \mapsto (a, a, a) - \mathbf{p}$, both map the frame and the line OO' to themselves. So they don't change any moment of inertia about OO'. The rotation carries OA to OB and OC, and the reflection carries these to $O'A'$, $O'B'$ and $O'C'$. These are the six rods with an end at O or O', and each has moment of inertia $2ma^2/9$. In the same way, $B'C$ is carried to the other five rods of the "zig-zag" AB', $B'C$, CA', $A'B$, BC', $C'A$ round the middle of the cube, and each of these has moment of inertia $5ma^2/9$. Hence

$$I_{\text{frame}} = 6 \cdot \frac{2ma^2}{9} + 6 \cdot \frac{5ma^2}{9} = \frac{42ma^2}{9} = \frac{14ma^2}{3},$$

as required.

Angular acceleration. Now the frame can only turn about the fixed line OO'. The bearings supply whatever couple is needed perpendicular to that line, so only the component of the applied couple along OO' turns the frame. The forces still form the couple $\mathbf{G} = amg(1, 1, 2)$, and its moment about any line is its component along that line:

$$\mathbf{G} \cdot \mathbf{u} = \frac{amg(1 + 1 + 2)}{\sqrt{3}} = \frac{4amg}{\sqrt{3}}.$$

The equation $I\ddot{\theta} = \mathbf{G} \cdot \mathbf{u}$ for rotation about a fixed axis gives

$$\ddot{\theta} = \frac{4amg/\sqrt{3}}{14ma^2/3} = \frac{12g}{14\sqrt{3}\,a} = \frac{2\sqrt{3}\,g}{7a}.$$

Its sense is anticlockwise when viewed looking from O' towards O, that is, a right-handed rotation about the direction $\overrightarrow{OO'}$.

Checks

- Each of the six forces has a "partner": OA with $C'B$, AC' with $A'C$, and $O'A$ with OA'. Each pair is equal and opposite on parallel lines, which is why the resultant is zero.
- The frame has cubic symmetry about its centre, so its moment of inertia is the same about every axis through the centre. About the axis through the centre parallel to OC, the four vertical rods contribute $4m(a/\sqrt{2})^2 = 2ma^2$. The eight horizontal rods each contribute $\frac{ma^2}{12} + \frac{ma^2}{4} = \frac{ma^2}{3}$. The total is $\frac{14ma^2}{3}$, agreeing with the answer above, since OO' also passes through the centre.
- Numerically, with $a = m = g = 1$, midpoint sums over the twelve rods give $I = 4.66667$ and $\ddot{\theta} = 0.494872$, matching $2\sqrt{3}/7 \approx 0.494872$.

Common slips

- Forgetting the factor $1/\sqrt{2}$ when converting the face-diagonal forces $\sqrt{2}\,mg$ into components: each has components of size mg, not $\sqrt{2}\,mg$.
- Using the full couple $|\mathbf{G}| = \sqrt{6}\,amg$ in part (ii) instead of its component along the fixed axis.
- Applying the parallel axis theorem from an axis that doesn't pass through the centre of mass: it must always go *via* the centre of mass, as in the step from $2ma^2/9$ to $ma^2/18$.

Beyond STEP

For rotation in three dimensions, the moment of inertia becomes a symmetric 3×3 matrix, the inertia tensor, with angular momentum $\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$. Its eigenvectors are the principal axes. As the note in part (i) shows, a body does not in general start to turn about the axis of the applied couple. The cube frame is a nice example: it has the same moment of inertia about every axis through its centre, but this is not true about a corner. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid body motion in three dimensions, including the inertia tensor.

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